MIL-V-0050010A(MU)
16 March 1959
USED IN LIEU OF
MIL-V-50010
22 May 1959

MILITARY SPECIFICATION

VALVE, ANTIBACK DRAFT, PROTECTIVE SHELTER, M2

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-V-50010. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of valve for protective shelters.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530	-	Plywood, Flat-Panel.
QQ-S-781	-	Steel, Strapping, Flat.
TT-W-00572	-	Wood-Preservative; Water-Repellent.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Box, Fiberboard.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

MILITARY

MIL-P-116	-	Preservation, Methods of.

FSC 4240

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-65-32 - Valve, Antiback Draft, Protective Shelter, M2.
D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations Inc, 1616 P Street, N. W. Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-65-32 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the antiback draft valve shall conform to the specifications and drawings listed on Drawing DL5-65-32 and to subsidiary drawings.

3.2 Manufacture and assembly. The antiback draft valve shall be manufactured and assembled in accordance with Drawing D5-65-32.

3.3 Closure. The gap between the valve cover and the felt wool gasket shall be no more than 1/32 inch with the valve in its operating position when tested as specified in 4.4.4.1.

3.4 Performance. The valve cover opening shall be 2.8 $\pm$ 0.3 inches between points A and B (see figure 1) with the pointer indicating 12° $\pm$ 2° on the gage inside the valve body when the valve is mounted in its operating position and subjected to an airflow of 400 cubic feet per minute (CFM) at a static pressure of 0.30 $\pm$ 0.03 inch of water, when tested as specified in 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of antiback draft valves shall be produced in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. The antiback draft valve shall be free from contamination (dirt, grease, oil, or foreign matter) and damage (dents, cracks, breaks, or deformations) which would impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of two antiback draft valves shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure for examination and nondestructive tests. The preproduction sample of antiback draft valves shall be examined in accordance with the classification of defects 4.4.3.2, and tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the antiback draft valves produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination and nondestructive tests. Sample antiback draft valves and the levels A and B preparation for delivery shall be examined and tested in accordance with the classification of defects (4.4.3.2) and MIL-STD-105.

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4.4.3.2 Classification of defects.

(a) Body, valve (Dwg D5-65-43).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Body width
102	Body height
103	Diameter of mounting holes in flange
104	Location of mounting holes in flange

(b) Valve, antiback draft, protective shelter, M2 (Dwg D5-65-32).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Closure	4.4.4.1
102	Performance	4.4.4.2
103	Overall dimensions	
104	Welding	
105	Counterweight not free to move the entire length of shaft	
106	Pointer not indicating zero on gage when valve is closed	
107	Workmanship (3.6)	
Minor:	AQL 4.0 percent defective	
201	Protective finish missing, inadequate, or incorrect	
202	Felt gasket not secure	

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4.4.3.2 Classification of defects (continued).

(c) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Contamination on unpainted surfaces
102	Preservative incorrect or missing where required
103	Container or container closure incorrect, torn, or broken
104	Fiberboard cells missing or incorrectly applied
105	Strapping loose or missing
106	Marking missing, incorrect, or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Closure. Set the antiback draft valve in the operating position (i.e., as if mounted in a vertical wall with the hinges at the top), and adjust the counterweight so that the valve cover is closed. Measure the widest gap (if any) between the cover and the felt wool gasket.

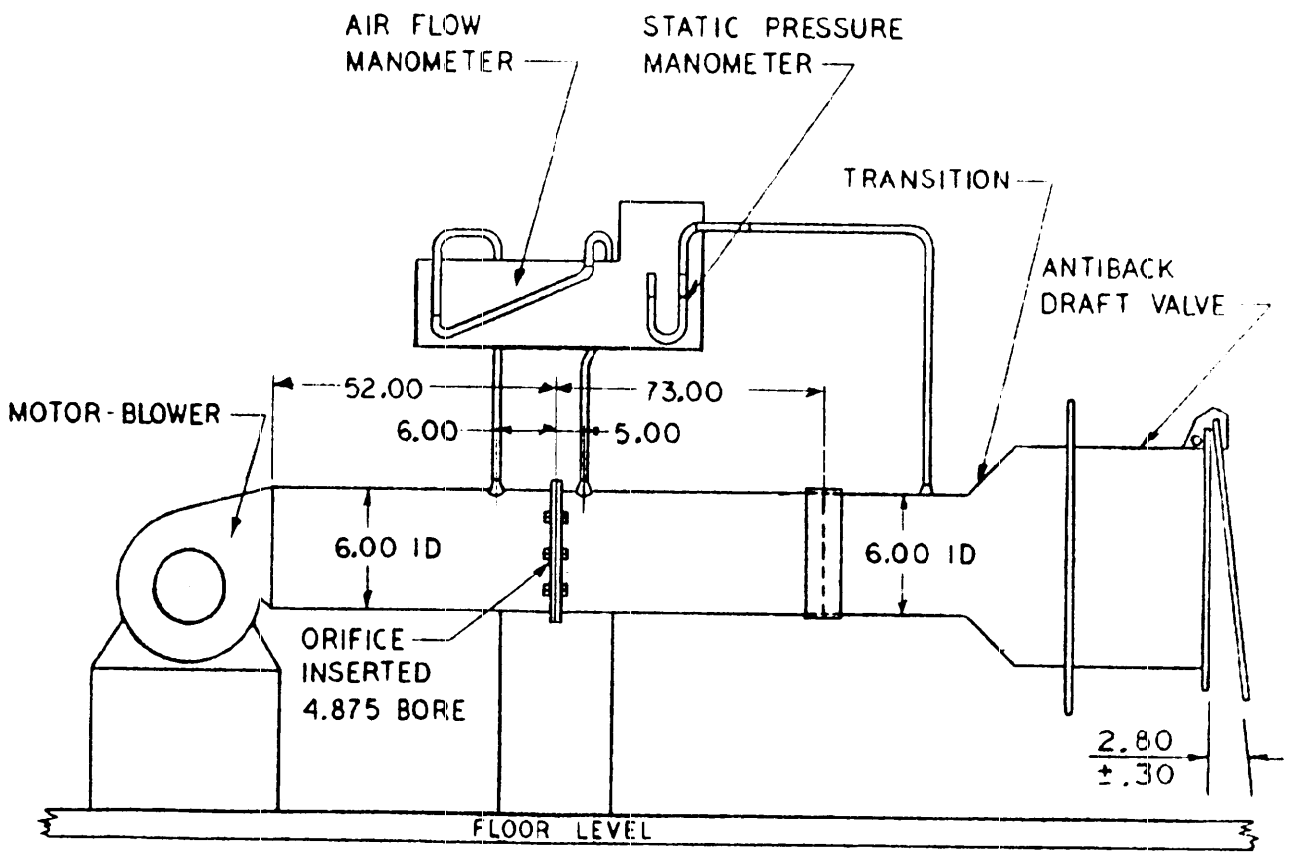
4.4.4.2 Performance. Connect the antiback draft valve to a suitable source of air as shown in figure 1. Turn on the air source and adjust to the specified airflow. Read the static pressure, check the angle on the gage inside the valve body indicated by the pointer, and measure the distance between points A and B (see figure 1).

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. Unpainted metal surfaces of the antiback draft valve, including threaded areas shall be cleaned and dried in accordance with MIL-P-116, using cleaning method C-1 and any of the drying methods listed therein.

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CODE 81361	PART OF SPEC MIL-V-0050010A	SPECIFICATION FIGURE DRAWING
		DATE 11 FEB 69
		136-301-39



NOTE: SILVER BRAZE ALL TAPS FLUSH WITH INSIDE OF DUCT

FIGURE 1

TEST APPARATUS FOR VALVE PERFORMANCE

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5.2 Preservation and packaging. Level shall be as specified (see 6.2).

5.2.1 Level A. All unpainted metal surfaces of the antiback draft valve, cleaned and dried as specified in 5.1, shall be coated with type P-2 preservative compound conforming to MIL-P-116. The valve cover shall be secured in a closed position by taping around the cover seam with class 2 pressure-sensitive tape conforming to PPP-T-60. The adjustable flange shall be bolted to the valve body near the hinge. The counterweight and shaft shall be secured to the flange as follows: Insert one end of the threaded shaft approximately 1-1/2 inches through the center hole in one end of the mounting flange and screw the counterweight down tightly against the mounting flange surface. Install two of the nuts on that portion of the threaded shaft protruding through the mounting flange tightly securing the shaft in place. Install the third nut on the opposite end of the shaft screwing it down tightly against the counterweight to prevent it from working loose (see Drawing D5-65-1261). To prevent movement of the counterweight arm, apply a piece of class 2 tape conforming to PPP-T-60 around the arm drawing the ends of tape tight and securing these ends to the outside of the valve body.

5.2.2 Level C. The antiback draft valve shall be packaged to provide adequate protection against corrosion, moisture, dust, fungus, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.3 Packing. Level shall be as specified (see 6.2).

5.3.1 Level A. The antiback draft valve, preserved and packaged as specified in 5.2.1, shall be packed in an overseas type, style A, cleated plywood box conforming to PPP-B-601. The plywood used in construction of the box shall comply with group A of NN-P-530 in accordance with type I, grade 3 or 4 of CS-35 provided the plywood does not have continuous holes or openings extending through the entire panel from face to back. The box shall be closed and strapped in accordance with the appendix of the box specification. Strapping conforming to type I, class B of QQ-S-781 shall be used. The boxes, or the finished wood parts thereof, shall be completely immersed for a minimum of three minutes in a water repellent wood preservative conforming to composition A of TT-W-00572.

5.3.2 Level B. The antiback draft valve, preserved and packaged as specified in 5.2.1, shall be packed in a V3c, style RSC fiberboard box conforming to PPP-B-636. The box shall be closed, taped, and reinforced in accordance with the appendix of PPP-B-636 for exterior boxes, using galvanized strapping in the box reinforcement.

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5.3.3 Level C. The antiback draft valve, preserved and packaged as specified in 5.2.2, shall be packed to insure carrier acceptance and safe delivery at first domestic destination for immediate use. Containers shall be in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing D150-2-1. All markings on the shipping container shall be clear, legible, and permanent after application of the wood preservative specified in 5.3.1.

6. NOTES

6.1 Intended use. The antiback draft valve is intended to automatically regulate the pressure and airflow rate between the outer air locks of protective shelters and the atmosphere.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see Section 5).
(c) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facilities and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 4240-A385

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel (either government or contractor) involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines of reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents(s) or to amend contractual requirements.

SPECIFICATION

MIL-V-005001A (MU)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER(S)

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED

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